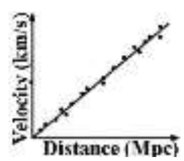
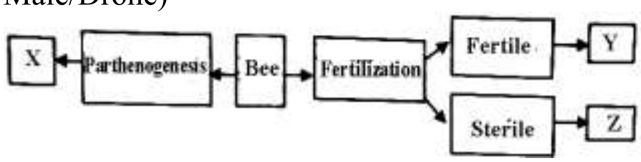
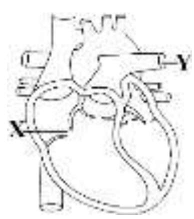
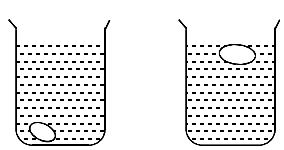
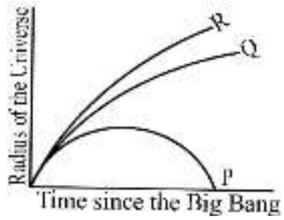
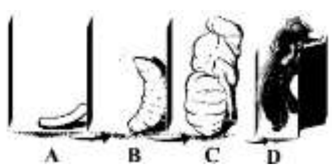
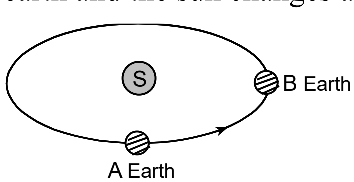
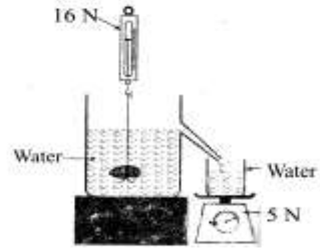
First Term Examination ~ 2083
Subject: Science and Technology
(Questions and Solutions)

Group A		
Q.N.	**Questions**	**Answers**
1. a.	**In the life cycle of honey bee, which stage of the following is the voracious eater?** i. Egg ii. larva iii. Pupa iv. Adult	ii. larva
1. b.	**Which valve controls the blood flow from right ventricle to right auricle?** i. Aortic valve ii. Bicuspid valve iii. Pulmonic valve iv. Tricuspid valve	iv. Tricuspid valve
1. c.	**What is the weight of the body on the surface of the moon, if its weight is 150 N on the earth surface.** i. 20 N ii. 25 N iii. 30 N iv. 35 N	ii. 25 N
1. d.	**What does the slope of the given graph represent?** i. Acceleration of universe ii. Velocity of universe iii. Time of origin of universe iv. Hubble's constant 	iv. Hubble's constant
1. e.	**Which of the following groups indicate the correct order of increasing metallic character of the elements?** i. Be < Mg < Ca ii. Na > Li > K iii. Mg < Al < Si iv. C < O < N	i. Be < Mg < Ca
1. f.	**What is X in the given chemical equation?** $H_2SO_4 + 2NaOH \rightarrow X + H_2O$ i. Na_2SO_3 ii. $NaSO_3$ iii. $NaSO_2$ iv. Na_2SO_4	iv. Na_2SO_4

Group B		
Q.N.	**Questions**	**Answers**
2. a.	What is nuptial flight or mating flight?	Nuptial flight is the mating flight of a queen bee with the drones in the open air.
2. b.	The blood pressure of a man is generally 120/80	It means the systolic blood pressure (maximum

	mmHg. What does it mean?	pressure during heart contraction) is 120 mmHg, and the diastolic blood pressure (minimum pressure during heart relaxation) is 80 mmHg.
2. c.	Write a difference between mass and weight.	Mass is the total amount of matter contained in a body (measured in kg), whereas weight is the gravitational force exerted by a heavenly body on an object (measured in N).
2. d.	Write the relationship between liquid density and buoyant force.	Liquid density is directly proportional to buoyant force.
2. e.	Write modern periodic law.	The modern periodic law states that the physical and chemical properties of elements are the periodic functions of their atomic numbers.
2. f.	Write the name of compound which is formed when ammonia dissolves in water.	Ammonium hydroxide (NH_4OH).
	Group C	
3.	Study the concept map and write the function of Y and Z. (Y: Fertile Female, Z: Sterile Male/Drone) 	**Y (Queen bee):** Main function of Queen bee is to mate and lay eggs. **Z (Drone bee):** Main function of drone bee is to mate with the queen bee.
4.	According to given diagram, write the function of X and Y. Write one effect due to lack of fibrinogen in blood. 	**Function of X:** X carries blood from right auricle to right ventricle. **Function of Y:** Y carries blood from heart to the different parts of the body. **Effect:** Blood will lose its ability to clot, leading to excessive bleeding from minor cuts.
5.	The mass and radius of Jupiter are 1.9×10^{27} kg and 71×10^{6} m respectively, find the acceleration due to gravity of it. What will be the weight of an object having mass 80 kg on that planet?	**g** $= GM/R^2 = (6.67 \times 10^{-11} \times 1.9 \times 10^{27}) / (71 \times 10^{6})^2 = 25.14\ m/s^2$. **Weight (W)** $= m \times g = 80 \times 25.14 = 2011.2$ N.
6.	In the given figure, which is sea water and which is river water? Why?  (A) (B)	**B is sea water** and **A is river water**. **Reason:** Sea water has dissolved salts, giving it a higher density. The egg floats higher in the denser liquid to balance its weight due to greater upthrust.
7.	By using a hydraulic lift a load of 2500 N needs to be lifted. If cross-sectional area of small piston is 1/10 of the big piston, calculate the force required to apply on small piston.	According to Pascal's law: $F_1/A_1 = F_2/A_2$ $F_1 = F_2 \times (A_1/A_2) = 2500 \times (1/10) =$ **250 N**.

8.	Study the graph of hypothesis about future of universe given aside. What does the curve R represent? In which respect P is different from R? 	Curve **R** represents an "Open Universe" (expanding forever). Curve **P** represents a "Closed Universe" which will eventually contract (Big Crunch), whereas **R** will expand indefinitely.
9.	Write the valency of the most reactive metal and the most reactive non-metal from the given table. **Na | Mg | Al | Si | P | S | Cl | Ar**	Most reactive metal: **Sodium (Na)**, Valency = 1. Most reactive non-metal: **Chlorine (Cl)**, Valency = 1.
10.	Write any two differences between open universe and closed universe.	1. An open universe will expand forever, while a closed universe will eventually stop expanding and contract. 2. An open universe has a density less than the critical density, whereas a closed universe has a density greater than the critical density.
11.	a. $CaCO_3 \rightarrow CaO + CO_2$ b. $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$ i. Identify the endothermic reaction. ii. What happens when powder of $CaCO_3$ is used?	i. Reaction **(a)** is endothermic as it requires heat. ii. Using powdered $CaCO_3$ increases the rate of the chemical reaction because of the increased surface area.
	Group D	
12.	i. Label A, B, C and D. ii. What is the process called when stage B sheds its old skin? iii. Under what condition does B develop into a worker bee? iv. How long does B take to form its next stage? 	i. A = Egg, B = Larva, C = Pupa, D = Adult. ii. Molting. iii. It develops into a worker bee if it is fed royal jelly for the first 3 days and then bee bread for the rest of the stage. iv. It takes about 5.5 to 6 days.
13.	Report of blood test... i. What could be the problem in him? ii. Write the name of blood cell 'B'. iii. Which disease may occur if number of blood cell 'C' is too low? iv. If he has a cut in his finger and bleeding does not stop, which one among A, B and C may have lesser than normal?	i. The person has a very low WBC count (3,000), indicating Leukopenia / weak immune system. ii. Platelets (Thrombocytes). iii. Anaemia. iv. Cell B (Platelets).
14.	a. Explain how the magnitude of the gravitational force between the earth and the sun changes as the earth moves from position 'A' to 'B'. b. Prove that: $g \propto 1/R^2$ 	**a.** As the earth moves from A to B, the distance between them decreases. Since gravitational force is inversely proportional to the square of the distance ($F \propto 1/d^2$), the force increases. **b.** $F = GMm/R^2$ (Newton's Law of Gravitation) and $F = mg$. Equating them: $mg = GMm/R^2 \Rightarrow g = GM/R^2$. Since G and M are constant, $g \propto 1/R^2$.

15.	A weight of piece of stone, when immersed in water is 16 N and it displaces 5 N water... i. What is the weight of stone in the air? ii. On which principle is the experiment based on? State this principle. iii. Calculate the mass of water displaced if 1 kg is equal to 10 N. 	i. Weight in air = Weight in water + Upthrust = 16 N + 5 N = 21 N. ii. **Archimedes' Principle**: When a body is fully or partially immersed in a fluid, it experiences an upthrust equal to the weight of the fluid displaced. iii. Mass (m) = Weight/g = 5 N / 10 N/kg = 0.5 kg.
16.	Study the part of periodic table... i. Which one of them is very active non-metal? ii. Write the electronic configuration of element 'W' on the basis of sub-shell. iii. Write the reason of being 'Y' more reactive than 'X'. iv. Write the balanced chemical equation between the reaction of Z and M.	i. **M** (Group 17, Chlorine) is the most active non-metal. ii. 'W' is Argon (Group 18, Period 3, atomic no. 18): $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$. iii. 'Y' (Sodium) is more reactive than 'X' (Lithium) because Y has a larger atomic radius, making it easier to lose its valence electron. iv. Z (Mg) and M (Cl) reaction: $Z + M_2 \rightarrow ZM_2$ ($Mg + Cl_2 \rightarrow MgCl_2$).

*** Best of Luck***